12.6 One More Coin

WORKSHEET

The question

STEP I 2016, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q12

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Alice tosses twice: what is the distribution of her number of heads? And Bob, tossing three times?

STAGE 2 Splitting Bob's tosses

Ask yourself Think of Bob's $n+1$ tosses as n tosses followed by one extra toss. After his first n tosses, he is ahead of Alice, level with her, or behind. What happens in each case?

Pause and try Here is the one-sentence proof. Bob has $n+1$ tosses, so his heads and tails add to $n+1$, one more than Alice's. Show that Bob has more heads than Alice if and only if he does *not* have more tails than Alice. Since heads and tails are symmetric, these two events have equal probability, so each is $\frac{1}{2}$.

What's really going on

Symmetry and complements

The slick proof combines two facts: two events that are swapped by a symmetry have equal probabilities, and two complementary events have probabilities adding to 1. If both hold, each has probability $\frac{1}{2}$. The same reasoning appears throughout probability. In the airline problem (Section 12.5), S_1 and S_n were swapped by a symmetry and exactly one of them is left at the end. In a game of Hex, the board's symmetry and the fact that exactly one player must win show that the first player has the advantage, though nobody knows a winning strategy on large boards.

How likely is a tie?

For equal numbers of tosses the tie probability p_1 matters. Alice and Bob tie when Alice's heads equal Bob's heads, which (replacing Bob's heads by his tails, by symmetry) has the same probability as Alice's heads plus Bob's heads equalling n: a total of n heads in $2n$ tosses. So

$$p_1 = \binom{2n}{n}\frac{1}{4^n} \approx \frac{1}{\sqrt{\pi n}},$$

using Stirling's approximation (Section 12.23). Ties become rare, but slowly: with 100 tosses each, the chance of a tie is still about 5.6%.

Try this next

1. **Two extra coins.** If Bob tosses $n+2$ coins and Alice n, prove $\mathrm{P}(\text{Bob more heads}) = \frac{1}{2} + \frac{1}{2}\mathrm{P}(H' = H_A)$, where H' is the number of heads in Bob's first $n+1$ tosses.
2. **Biased coins.** If both use a coin with probability p of heads, is the answer still $\frac{1}{2}$ with one extra toss? Which step of the slick proof fails?
3. **Ties.** Show that $p_1 = \binom{2n}{n}/4^n$ by the trick in the text, and evaluate it for $n = 1, 2, 3$.
4. **Dice.** Alice rolls n dice and Bob rolls $n+1$. Is the probability that Bob gets more sixes equal to $\frac{1}{2}$?

Further reading

- **Peter Winkler, *Mathematical Mind-Benders*** (A K Peters). Many probability puzzles with symmetric solutions.
- **Gabor Székely, *Paradoxes in Probability Theory and Mathematical Statistics*** (Reidel).
- **Frederick Mosteller, *Fifty Challenging Problems in Probability*** (Dover).

The full guided solution is in Section 12.6:
https://beyondstep.keithmcnulty.org/ch12/one-more-coin.html